

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain Ionic bond, Covalent bond, Metallic bond in detail.	L2	6
B) Explain 1) Drift velocity 2) Mobility 3) Collision time 4) Mean free path 5) Relaxation time	L2	6
C) Explain concept of Electrical conductivity in detail.	L2	6
Q.2 Solve Any Two of the following.		12
A) Explain Joule's Law in detail.	L2	6
B) Explain Superconductivity and superconductor and application of superconductor	L2	6
C) What is polarization? Explain all types of polarization	L2	6
Q. 3 Solve Any Two of the following.		12
A) What is semiconductor? and Write properties of semiconductor	L3	6
B) Write note on Fermi level and electron-hole concentration	L3	6
C) Draw and explain the bond diagram of semiconductor. And write application of semiconductor.	L3	6
Q.4 Solve Any Two of the following.		12
A) Write classification of magnetic materials and explain every type.	L3	6
B) Explain soft and hard magnetic materials	L3	6
C) Write properties of ferromagnetic materials in detail.	L3	6
Q. 5 Solve Any Two of the following.		12
A) Why galvanization and impregnation of materials is needed for materials	L3	6
B) What is refractory material? Explain properties of refractory materials.	L3	6
C) What is radioactive material? And applications of radioactive material.	L3	6

*** End ***